

# A Biface-Genesis Branch of Ghost-Free Bigravity

The Hassan–Rosen Origin of the BPB/MBE Effective Dictionary

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## Abstract

We introduce BPB/MBE as an effective dictionary generated by a late-time cosmological branch of ghost-free Hassan–Rosen bimetric gravity, rather than as a collection of independent phenomenological corrections. We call this trajectory the *biface-genesis branch*. Its branch variable is the scale-factor ratio  $y = a_f/a_g$ , and its effective descendants are obtained from the HR interaction structure evaluated along this branch. A central distinction is made between the action-level HR potential  $U_{\text{HR}}(X, y)$  and the  $g$ -Friedmann source polynomial  $V_g^{\text{F}}(y)$ . The permanent acceleration component is selected by the lapse-normalized action projection

$$\Omega_{\text{de}} = \Pi_{\text{perm}}[U_{\text{HR}}] \equiv U_{\text{HR}}(1, 0) = \beta_0 + \beta_1,$$

whereas  $V_g^{\text{F}}(y_t)$  fixes the transition branch density. With the rounded HR branch values, the lower Bianchi point is  $y_t = 0.300$ ,  $V_g^{\text{F}}(y_t) = 0.810$ ,  $\beta_0 + \beta_1 = 0.648$ , and the structural split is  $\Delta\Omega = 0.162 = y_t^2(3 - 4y_t)$ . The corresponding rounded HR survival value is

$$s_{\text{surv,HR}} = \frac{0.648}{0.810} = 0.800.$$

For numerical corpus locks and downstream applications we keep the independently source-locked value explicit,

$$s_{\text{surv},0} = 0.7956924397708606, \quad 1 - s_{\text{surv},0} = 0.2043075602291394.$$

This separation is intentional: the rounded HR algebra explains the branch mechanism, while the exact source-locked value is the reproducible numerical descendant used by the application papers. The normalized HR  $y$ -mode at the lower Bianchi point gives  $m_{\text{FP}} = 2.165824 H_0$  in the documented D2 convention bundle, and the Unruh–de Sitter bridge gives

$$a_0^{\text{BPB}} = cH_0(1 - s_{\text{surv},0}) = 1.367856592595 \times 10^{-10} \text{ m s}^{-2}$$

for the primary  $H_0 = 68.910590682276 \text{ km s}^{-1} \text{ Mpc}^{-1}$  lock. The raw value  $V_g^{\text{F}}(0) = \beta_0 \simeq 0.918$  is not the permanent component and must not be used as the numerator of a survival fraction. The same branch also contains a curvature threshold  $y_{\text{crit}} = -\beta_2/\beta_3$ , where  $V_g^{\text{F}''}(y)$  changes sign. The late-acceleration equation-of-state diagnostic is obtained directly from  $V_g^{\text{F}}(y)$  through  $y(z) = y_t(1 + z)$ ; it predicts  $w(0) = -1$ , a small positive local slope  $w_a \simeq 0.0667$ , and a second  $w = -1$  crossing at  $z = 2$ . The present branch value satisfies  $y_0 > y_{\text{crit}}$ , placing the late-time solution on the negative-curvature side and fixing the sign of the effective growth-drag descendant,  $\beta_{\text{grow}} < 0$ . Its magnitude is controlled, at the dictionary level, by the survival deficit  $1 - s_{\text{surv},0}$  and by a background growth correction,  $\beta_{\text{grow}} \simeq -(1 - s_{\text{surv},0})G_{\text{corr}}$ . The  $f$ -sector coefficient  $\beta_4$  is kept as a residual closure parameter; no condition  $V_f^{\text{F}}(y_t) = 0$  is imposed in the minimal dictionary. We formulate the resulting perturbative dictionary, state its EFT closure status, and give the corresponding parameter-counting rule for AIC/BIC comparisons. The observational applications are released as companion manuscripts; Paper 0 supplies their common upstream dictionary and source-locked bookkeeping. The present v0.9 release also points to the corpus-level audit artefacts, in particular `00_README_CORPUS`, `00_MANIFEST_CORPUS`, `00_CLAIMS_INDEX`, and `00_REVIEWER_ROUTE`, which separate orientation, file provenance, claim provenance, and referee navigation.

| Object                | Role in the Paper 0 dictionary                                                                                                     |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------|
| $U_{\text{HR}}(X, y)$ | Action-level HR interaction potential on the FLRW square-root spectrum.                                                            |
| $V_g^{\text{F}}(y)$   | $g$ -Friedmann source polynomial; not identical to the permanent action projection.                                                |
| $\Omega_{\text{de}}$  | Permanent branch component, $\Pi_{\text{perm}}[U_{\text{HR}}] = U_{\text{HR}}(1, 0) = \beta_0 + \beta_1$ .                         |
| $\Delta\Omega$        | Structural split, $V_g^{\text{F}}(y_t) - (\beta_0 + \beta_1)$ .                                                                    |
| $s_{\text{surv},0}$   | Survival descendant. Rounded HR value 0.800; exact corpus lock 0.7956924397708606.                                                 |
| $a_0^{\text{BPB}}$    | Acceleration-scale descendant, $cH_0(1 - s_{\text{surv},0}) = 1.367856592595 \times 10^{-10} \text{ m s}^{-2}$ .                   |
| $m_{\text{FP}}$       | Closure-selected HR mode mass, $2.165824 H_0$ , associated with the D2 acceleration-scale lock.                                    |
| $y_{\text{crit}}$     | Curvature threshold, $-\beta_2/\beta_3$ .                                                                                          |
| $\beta_{\text{grow}}$ | Growth-drag descendant, with sign fixed by $y_0 > y_{\text{crit}}$ and magnitude set by $(1 - s_{\text{surv},0})G_{\text{corr}}$ . |
| $\beta_4$             | Residual $f$ -sector/EFT closure parameter.                                                                                        |

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# 1 Introduction

The purpose of Paper 0 is to define the theoretical dictionary of BPB/MBE from ghost-free Hassan–Rosen bimetric gravity. It is the entry point of a corpus release, not an isolated phenomenological note. Its role is to identify the HR branch, define the branch locks, derive the effective descendants that enter the companion papers, and state how those descendants should be counted in model comparison. It is not, by itself, a global observational likelihood paper and it does not attempt to close every perturbative or non-linear sector.

The central claim is that BPB/MBE is not a collection of independent phenomenological patches. It is an effective dictionary generated by a late-time HR cosmological trajectory, which we call the *biface-genesis branch*. This branch is described by the scale-factor ratio

$$y(a) = \frac{a_f}{a_g}, \quad (1)$$

and by the HR interaction structure evaluated along that trajectory.

A key point of the present version is the distinction between two quantities that are easily conflated. The action-level interaction potential evaluated on a FLRW square-root matrix  $S = \text{diag}(X, y, y, y)$  is denoted  $U_{\text{HR}}(X, y)$ . The  $g$ -Friedmann source polynomial obtained after varying the lapse is denoted  $V_g^{\text{F}}(y)$ . They are related, but they are not the same projection of the HR interaction. Therefore

$$V_g^{\text{F}}(0), \quad \Pi_{\text{perm}}[U_{\text{HR}}], \quad V_g^{\text{F}}(y_t) \quad (2)$$

are distinct objects.

The branch-projection rule used here is

$$\Omega_{\text{de}} = \Pi_{\text{perm}}[U_{\text{HR}}] = U_{\text{HR}}(1, 0) = \beta_0 + \beta_1. \quad (3)$$

Thus the permanent acceleration component is not introduced as the raw  $g$ -Friedmann value  $V_g^{\text{F}}(0) = \beta_0$ . In the rounded branch lock,

$$\beta_0 \simeq 0.918, \quad \beta_1 \simeq -0.270, \quad \beta_0 + \beta_1 = 0.648. \quad (4)$$

The rounded HR survival value is consequently

$$s_{\text{surv,HR}} = \frac{\beta_0 + \beta_1}{V_g^{\text{F}}(y_t)} = \frac{0.648}{0.810} = 0.800, \quad (5)$$

not  $V_g^{\text{F}}(0)/V_g^{\text{F}}(y_t) = 0.918/0.810 > 1$ . The exact numerical value propagated by the current corpus and used by the application papers is kept as a separate source lock,

$$s_{\text{surv},0} = 0.7956924397708606, \quad 1 - s_{\text{surv},0} = 0.2043075602291394. \quad (6)$$

This distinction is important. The rounded HR algebra explains the mechanism and fixes the bookkeeping; the exact source-locked value fixes the reproducible numerical descendant used in later gates.

The second descendant is the effective growth-drag coefficient  $\beta_{\text{grow}}$ . Its sign is tied to the negative-curvature side of the branch, defined by

$$y_{\text{crit}} = -\frac{\beta_2}{\beta_3}, \quad (7)$$

where  $V_g^{\text{F}''}(y_{\text{crit}}) = 0$ . The BPB/MBE trajectory satisfies  $y_0 > y_{\text{crit}}$  today in the branch solution. The sign of  $\beta_{\text{grow}} < 0$  is therefore structural. Its magnitude is set, at the effective-dictionary level, by the survival deficit  $1 - s_{\text{surv},0}$  and a background growth correction  $G_{\text{corr}}$ .

The v0.9 update aligns Paper 0 with the locked downstream corpus, adds explicit root-index navigation, and states the release architecture explicitly: the D2 acceleration-scale bridge is stated as source-locked under the documented H1–H4 convention bundle, and the acceleration-scale descendant is updated to

$$a_0^{\text{BPB}} = cH_0(1 - s_{\text{surv},0}) = 1.367856592595 \times 10^{-10} \text{ m s}^{-2} \quad (8)$$

for  $H_0 = 68.910590682276 \text{ km s}^{-1} \text{ Mpc}^{-1}$ . This does not turn Paper 0 into an observational fit; it makes the dictionary numerically traceable and explains why the companion papers exist.

The paper is organized as follows. Section 3 recalls the ghost-free HR action and introduces the action-level potential. Section 4 specializes it to FLRW and distinguishes  $U_{\text{HR}}(X, y)$  from  $V_g^{\text{F}}(y)$ . Section 5 states the branch locks and the branch-projection rule. Section 6 gives the dynamical branch-selection interpretation. Section 7 identifies the tachyonic-response threshold. Sections 8, 9, and 10 derive the survival, acceleration-scale, and growth descendants. Section 11 states the perturbation dictionary and EFT closure status. Section 12 discusses AIC/BIC parameter counting. Section 13 gives the effective late-acceleration equation-of-state diagnostic. Section 2 states the corpus-release architecture. Sections 14–15 separate companion applications from the claims of Paper 0.

## 2 Corpus release architecture and companion papers

Paper 0 is designed to be read as the foundational dictionary of a structured release. The companion manuscripts are not assumptions used to derive the branch. They are downstream applications of the same source-locked dictionary. The logical direction is therefore

$$\text{HR branch} \longrightarrow \text{BPB/MBE dictionary} \longrightarrow \text{sector-specific tests.} \quad (9)$$

This direction is essential. If the companion sectors are hidden, the Paper 0 derivation can look unnecessarily abstract; if they are used as premises, the dictionary becomes circular. The release therefore separates the roles: Paper 0 derives and audits the dictionary, while Papers I–VII test different descendants of that dictionary.

| Companion | Sector                                            | Paper 0 descendant used                                                                                   | Role |
|-----------|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------|------|
| Paper I   | Low-redshift background, growth, and lensing      | $s_{\text{surv},0}$ , $\beta_{\text{grow}}$ , and the distinction between matter growth and Weyl response | Test |
| Paper II  | SPARC rotation curves and galaxy-scale amplitudes | Projected Weyl/matter kernel and source-side amplitude response                                           | Test |
| Paper III | CMB and lensing-sector response                   | Survival/face-minus branch response and Weyl-sector closure                                               | Test |
| Paper IV  | Compact objects / neutron stars                   | Strict dictionary projection in dense matter and tidal response                                           | Test |
| Paper V   | Early galaxies                                    | Survival-normalized abundance and high-redshift mapper consequences                                       | Test |
| Paper VI  | $E_G$ and Weyl–matter diagnostics                 | $r_{W\delta} = \sqrt{a_{\text{bg}}}$ and density-weighted projector structure                             | Test |
| Paper VII | BTFR and acceleration scale                       | $a_0^{\text{BPB}} = cH_0(1 - s_{\text{surv},0})$ and the D2 thermal bridge                                | Test |

This is not a claim that all companion sectors have the same maturity. The corpus uses explicit status labels: source-locked identities, mechanism gates, raw diagnostic gates, likelihood gates, and guardrails are kept distinct. In particular, a companion-paper pass does not promote every sector to a full final likelihood unless that likelihood is explicitly part of the corresponding gate.

The corpus root contains four non-scientific but audit-critical navigation files. They do not add claims to the model; they define how a reader, referee, or reproducer should locate the claims and their source locks:

| Root artefact            | Function                                                                                                       | Status    |
|--------------------------|----------------------------------------------------------------------------------------------------------------|-----------|
| 00_README_CORPUS.pdf     | Human-readable entry point: series map, claims discipline, and guardrail summary.                              | Orienting |
| 00_MANIFEST_CORPUS.csv   | Machine-readable file map for the corpus layers, manuscript/source paths, source-lock paths, and next actions. | Index     |
| 00_CLAIMS_INDEX.csv      | Normalized list of source-locked claims harvested from paper/source-lock outputs.                              | Index     |
| 00_REVIEWER_ROUTE.md/csv | Short route for a skeptical reader: where to verify or falsify the main claims.                                | Route     |

These root artefacts are intentionally separated from the scientific derivation. A claim is not valid because it appears in the manifest; it is valid only to the extent that the corresponding paper and source-lock artefact support it. The manifest and claims index are therefore reproducibility infrastructure, not additional evidence.

The practical consequence for reading the release is simple. Paper 0 should be cited for the HR origin, branch projection, survival factor, growth-drag sign, acceleration-scale identity, and parameter-counting rule. The companion papers should be cited for the observational or astrophysical performance of each descendant. This prevents the corpus from becoming either a single overgrown manuscript or a set of disconnected phenomenological patches.

### 3 Ghost-free Hassan–Rosen bigravity

The starting point is the ghost-free bimetric theory introduced by Hassan and Rosen as the dynamical-metric completion of the ghost-free massive-gravity potential. The construction removes the Boulware–Deser ghost by using a special interaction built from elementary symmetric polynomials of a matrix square root [4–7].

#### 3.1 Action

We write the bimetric action as

$$S = \frac{M_g^2}{2} \int d^4x \sqrt{-g} R[g] + \frac{M_f^2}{2} \int d^4x \sqrt{-f} R[f] - m^2 M_g^2 \int d^4x \sqrt{-g} \sum_{n=0}^4 \beta_n e_n \left( \sqrt{g^{-1}f} \right) + S_m[g, \psi]. \quad (10)$$

Here  $g_{\mu\nu}$  is the physical metric coupled minimally to matter,  $f_{\mu\nu}$  is the second dynamical metric, and  $e_n$  are the elementary symmetric polynomials of the eigenvalues of

$$S^\mu{}_\nu = \left( \sqrt{g^{-1}f} \right)^\mu{}_\nu. \quad (11)$$

The coefficients  $\beta_n$  are dimensionless interaction parameters.

#### 3.2 Action-level interaction potential

On a bimetric FLRW ansatz the square-root matrix has eigenvalues

$$S^\mu{}_\nu = \text{diag}(X, y, y, y), \quad X \equiv \frac{N_f}{N_g}, \quad y \equiv \frac{a_f}{a_g}. \quad (12)$$

The action-level HR interaction polynomial is then

$$U_{\text{HR}}(X, y) = \beta_0 + \beta_1(X + 3y) + \beta_2(3Xy + 3y^2) + \beta_3(3Xy^2 + y^3) + \beta_4Xy^3. \quad (13)$$

This object is the natural place to define the permanent branch projection, because it keeps the lapse eigenvalue  $X$  visible. The  $g$ -Friedmann source polynomial used below is a different projection, obtained through the lapse variation. The branch-projection rule used in this paper is

$$\Pi_{\text{perm}}[U_{\text{HR}}] \equiv U_{\text{HR}}(1, 0) = \beta_0 + \beta_1. \quad (14)$$

The choice  $X = 1$  is the lapse-normalized permanent projection of the action-level branch. It is a dictionary rule to be closed within the final HR closure; it is not the statement that the raw Friedmann source at  $y = 0$  equals the permanent density.

### 3.3 Matter coupling and assumptions

For the Paper 0 dictionary we assume single-metric matter coupling,

$$S_m = S_m[g, \psi], \quad (15)$$

so that the physical matter stress-energy tensor is conserved with respect to the  $g$ -metric connection. The  $f$  metric carries no independent matter fluid. This assumption keeps the dictionary conservative: all BPB/MBE descendants are traced to the HR interaction potential and the branch trajectory, not to an additional matter coupling.

## 4 FLRW ansatz and branch variable

### 4.1 Bimetric FLRW ansatz

We specialize Eq. (10) to homogeneous and isotropic backgrounds,

$$ds_g^2 = -N_g(t)^2 dt^2 + a_g(t)^2 \gamma_{ij} dx^i dx^j, \quad (16)$$

$$ds_f^2 = -N_f(t)^2 dt^2 + a_f(t)^2 \gamma_{ij} dx^i dx^j, \quad (17)$$

where  $\gamma_{ij}$  is the maximally symmetric spatial metric. The branch variable is

$$y(t) = \frac{a_f(t)}{a_g(t)}. \quad (18)$$

In HR cosmology this ratio determines the algebraic or dynamical branch structure of the solutions [8]. In the present paper it is the central coordinate of the BPB/MBE dictionary.

### 4.2 Friedmann-source polynomials

The action-level potential  $U_{\text{HR}}(X, y)$  should be distinguished from the polynomial that appears in the unprojected  $g$ -Friedmann equation after the lapse variation. We denote the latter by

$$V_g^{\text{F}}(y) = \beta_0 + 3\beta_1 y + 3\beta_2 y^2 + \beta_3 y^3. \quad (19)$$

The corresponding  $f$ -sector Friedmann-source polynomial is

$$V_f^{\text{F}}(y) = \beta_4 y^3 + 3\beta_3 y^2 + 3\beta_2 y + \beta_1. \quad (20)$$

A standard schematic form of the two Friedmann equations is

$$3H_g^2 = \frac{\rho_m}{M_g^2} + m^2 V_g^{\text{F}}(y), \quad (21)$$

$$3H_f^2 = m^2 \frac{V_f^{\text{F}}(y)}{y^3}. \quad (22)$$

The Bianchi identity imposes the propagation of  $y(a_g)$ . In a lapse gauge where  $N_g = 1$ , this may be written schematically as

$$\frac{dy}{d \ln a_g} = y \left( \frac{H_f}{H_g} - 1 \right), \quad (23)$$

up to the conventional lapse factor.

### 4.3 Action projection versus Friedmann source

The Paper 0 dictionary uses two HR projections:

$$\Omega_{\text{de}} = \Pi_{\text{perm}}[U_{\text{HR}}] = U_{\text{HR}}(1, 0) = \beta_0 + \beta_1, \quad (24)$$

$$\Omega_{\text{branch}}(y_t) = V_g^{\text{F}}(y_t). \quad (25)$$

This is the essential branch-projection distinction. The raw Friedmann-source value at  $y = 0$  is

$$V_g^{\text{F}}(0) = \beta_0, \quad (26)$$

whereas the permanent branch projection is

$$U_{\text{HR}}(1, 0) = \beta_0 + \beta_1. \quad (27)$$

Therefore  $V_g^{\text{F}}(0)$  need not equal  $\Omega_{\text{de}}$ . In the branch lock the difference is physical bookkeeping of the transient biface offset,

$$V_g^{\text{F}}(0) - \Omega_{\text{de}} = \beta_0 - (\beta_0 + \beta_1) = -\beta_1 \simeq 0.270. \quad (28)$$

### 4.4 Interpretation for BPB/MBE

The BPB/MBE dictionary is not built from arbitrary functions of time. It is built from the trajectory  $y(a_g)$  and from the HR polynomials in Eqs. (13), (19), and (20), with the  $f$ -sector normalization kept as a closure layer. The late-time branch selected by the BPB/MBE locks is the biface-genesis branch. Its role is to generate effective quantities that may later be projected onto CMB, growth, lensing, galactic, and compact-object observables.

## 5 BPB/MBE branch locks

The HR action contains five interaction coefficients  $\beta_0, \dots, \beta_4$ . The minimal Paper 0 dictionary fixes the part of this freedom relevant for the  $g$ -sector cosmological branch by imposing an action-level permanent projection, a transition branch-density lock, a stationary condition, and a curvature-threshold normalization. These are branch locks, not a full closure of every HR degree of freedom. In particular, the  $f$ -sector coefficient  $\beta_4$  remains a residual closure parameter.

### 5.1 Permanent projection and transition density

The permanent component is defined by the branch-projection rule

$$\Omega_{\text{de}} = \Pi_{\text{perm}}[U_{\text{HR}}] = U_{\text{HR}}(1, 0) = \beta_0 + \beta_1. \quad (29)$$

The transition branch density is defined in the  $g$ -Friedmann source projection,

$$V_g^{\text{F}}(y_t) = \Omega_{\text{de}} + \Delta\Omega. \quad (30)$$

For the branch values used here,

$$\Omega_{\text{de}} = 0.648, \quad V_g^{\text{F}}(y_t) = 0.810, \quad \Delta\Omega = 0.162. \quad (31)$$

The raw value  $V_g^{\text{F}}(0) = \beta_0 \simeq 0.918$  is not the permanent component. The ratio

$$\frac{V_g^{\text{F}}(0)}{V_g^{\text{F}}(y_t)} = \frac{0.918}{0.810} > 1 \quad (32)$$

is retained only as a consistency guardrail: it is not a survival fraction.

## 5.2 Minimal locks

The branch locks used in the minimal Paper 0 dictionary are the following. First, the permanent action-level projection is

$$\Pi_{\text{perm}}[U_{\text{HR}}] = U_{\text{HR}}(1, 0) = \beta_0 + \beta_1 = \Omega_{\text{de}}. \quad (33)$$

Second, the total transition branch normalization in the  $g$ -Friedmann source is

$$V_g^{\text{F}}(y_t) = \Omega_{\text{de}} + \Delta\Omega. \quad (34)$$

Third, the transition point is stationary in the  $g$ -sector branch polynomial,

$$V_g^{\text{F}'}(y_t) = 0. \quad (35)$$

Fourth, the curvature-threshold normalization is

$$y_{\text{crit}} = -\beta_2/\beta_3 \simeq 0.600. \quad (36)$$

With  $\beta_3 = -1$ , this fixes  $\beta_2 \simeq 0.600$ . Finally,  $\beta_4$  is not fixed at Paper 0 level. It remains an  $f$ -sector normalization and closure parameter, and it is not fixed by imposing  $V_f^{\text{F}}(y_t) = 0$ .

There is therefore no condition of the form  $V_g^{\text{F}}(0) = \Omega_{\text{de}}$  and no minimal lock  $V_f^{\text{F}}(y_t) = 0$ .

## 5.3 Branch solution for the $g$ -sector coefficients

With

$$y_t = 0.300, \quad y_{\text{crit}} = 0.600, \quad \beta_3 = -1, \quad (37)$$

the curvature lock gives

$$\beta_2 = 0.600. \quad (38)$$

The stationary lock  $V_g^{\text{F}'}(y_t) = 0$  gives

$$3\beta_1 + 6\beta_2 y_t + 3\beta_3 y_t^2 = 0 \quad \Rightarrow \quad \beta_1 = -0.270. \quad (39)$$

The permanent projection lock then gives

$$\beta_0 = \Omega_{\text{de}} - \beta_1 = 0.918. \quad (40)$$

Finally,

$$V_g^{\text{F}}(y_t) = 0.810, \quad (41)$$

so that

$$\Delta\Omega = V_g^{\text{F}}(y_t) - \Omega_{\text{de}} = 0.162. \quad (42)$$

Equivalently, under the current geometric normalization one may write

$$\Delta\Omega = y_t^2(3 - 4y_t) = 0.09 \times 1.8 = 0.162, \quad (43)$$

which shows that the split is tied to the branch geometry rather than inserted after the fact.



## 5.4 Why $V_f^F(y_t) = 0$ is not imposed

With the branch  $g$ -sector values,

$$3\beta_3 y_t^2 + 3\beta_2 y_t + \beta_1 = 0. \quad (44)$$

Thus

$$V_f^F(y_t) = \beta_4 y_t^3. \quad (45)$$

Imposing  $V_f^F(y_t) = 0$  would force  $\beta_4 = 0$  for  $y_t \neq 0$ . This contradicts the intended status of  $\beta_4$  as a residual  $f$ -sector closure parameter. The minimal Paper 0 dictionary consequently keeps  $\beta_4$  separate from the  $g$ -sector derivation of  $s_{\text{surv},0}$  and the curvature threshold.

## 5.5 Effective split and descendants

The split  $\Delta\Omega$  is a structural branch parameter. It is not identical to  $s_{\text{surv},0}$  or  $\beta_{\text{grow}}$ . Rather, it generates them through

$$\Delta\Omega \longrightarrow s_{\text{surv},0} = \frac{\beta_0 + \beta_1}{V_g^F(y_t)} = \frac{\Omega_{\text{de}}}{\Omega_{\text{de}} + \Delta\Omega} \longrightarrow \beta_{\text{grow}} \simeq -(1 - s_{\text{surv},0}) G_{\text{corr}}. \quad (46)$$

This is the first key parameter-counting point of the paper:  $s_{\text{surv},0}$  and  $\beta_{\text{grow}}$  are descendants of the HR branch dictionary, not additional independent inputs. Section 6 strengthens this interpretation by showing that the same lower Bianchi point can be read as the stationary endpoint of the reduced branch flow.

## 5.6 Reference branch values

The branch numerical target used in the BPB/MBE dictionary is

$$y_{\text{crit}} \simeq 0.600, \quad y_0 \simeq 0.670, \quad s_{\text{surv},0}^{\text{HR}} \simeq 0.800, \quad \beta_{\text{grow}}^{\text{HR}} \simeq -0.2067. \quad (47)$$

The exact numerical source lock used by the current corpus is

$$s_{\text{surv},0} = 0.7956924397708606, \quad 1 - s_{\text{surv},0} = 0.2043075602291394, \quad (48)$$

with the effective/MAP growth comparison  $\beta_{\text{grow}}^{\text{eff}} \simeq -0.2107$ . These numbers are not fitted inputs in the HR branch derivation; they are the reproducible numerical locks used when the dictionary is propagated into application papers.

# 6 Dynamical selection of the present branch

The previous section stated the BPB/MBE locks as algebraic branch conditions. There is a stronger and more physical reading of the same structure. The present value  $y_t = 0.300$  can be viewed as the stationary endpoint selected by the reduced Bianchi flow of the biface branch.

From Eq. (23), the scale-factor ratio obeys schematically

$$\frac{d \ln y}{d \ln a_g} = \frac{H_f}{H_g} - 1. \quad (49)$$

In the late branch used by the dictionary, the local relation is

$$y(z) = y_t(1 + z) = \frac{y_t}{a_g}, \quad (50)$$

so that  $d \ln y / d \ln a_g = -1$ . In the reduced branch-flow gauge this corresponds to a stationary  $f$ -sector endpoint,  $H_f/H_g = 0$ , at the present point. Equivalently, the branch selects a Bianchi stationary point rather than an arbitrary boundary value.

For the  $g$ -sector lock used here, the Bianchi polynomial entering the stationary condition is

$$\mathcal{B}(y) = \beta_1 + 2\beta_2 y + \beta_3 y^2. \quad (51)$$

Using

$$\beta_1 = -0.270, \quad \beta_2 = 0.600, \quad \beta_3 = -1, \quad (52)$$

one obtains

$$\mathcal{B}(y) = -(y - 0.300)(y - 0.900). \quad (53)$$

The two Bianchi points are therefore

$$y_{B1} = 0.300, \quad y_{B2} = 0.900. \quad (54)$$

The present branch chooses the lower stationary point,

$$y_t = y_{B1} = 0.300. \quad (55)$$

This gives a dynamical interpretation of the same condition written in Sec. 5 as  $V_g^{\text{F}'}(y_t) = 0$ .

The structural split then follows from the HR projections alone:

$$V_g^{\text{F}}(y_t) = 0.810, \quad (56)$$

$$\Pi_{\text{perm}}[U_{\text{HR}}] = U_{\text{HR}}(1, 0) = \beta_0 + \beta_1 = 0.648, \quad (57)$$

$$\Delta\Omega = V_g^{\text{F}}(y_t) - \Pi_{\text{perm}}[U_{\text{HR}}] = 0.162. \quad (58)$$

Thus  $\Delta\Omega$  is not introduced as a fitted late-time nuisance. It is the offset between the transition Friedmann source and the permanent action-level projection at the dynamically selected lower Bianchi point.

This statement should not be confused with imposing the stronger  $f$ -sector closure  $V_f^{\text{F}}(y_t) = 0$  as an additional Paper 0 lock. In the minimal dictionary  $\beta_4$  remains a residual  $f$ -sector closure parameter. The  $H_f = 0$  language used above refers to the reduced branch-flow interpretation of the present stationary endpoint; a fully source-matched  $f$ -sector background closure would require an additional convention for  $\beta_4$  and is not claimed here.

## 6.1 Chronology of the branch

The same branch gives a useful qualitative chronology. The entries in Table 1 are not additional observational inputs; they are landmarks of the HR polynomial branch.

| Branch region                    | Meaning                                                                                   | Status                      |
|----------------------------------|-------------------------------------------------------------------------------------------|-----------------------------|
| $y > y_{\text{bounce}}$          | Analytic continuation of the contracting $g$ -sector branch, with $V_g^{\text{F}} < 0$ .  | Open pre-bounce sector      |
| $y_{\text{bounce}} \simeq 1.647$ | First positive zero of $V_g^{\text{F}}$ ; candidate regular turnaround of the $g$ sector. | Derived polynomial landmark |
| $1.647 > y > 1$                  | Post-turnaround branch before metric-ratio symmetry.                                      | Perturbations open          |
| $y = 1$                          | Ratio-symmetry point $a_f = a_g$ .                                                        | HR branch landmark          |
| 0.900                            | Upper Bianchi point.                                                                      | Derived stationary point    |
| 0.300                            | Lower Bianchi point; present branch endpoint.                                             | Present lock                |
| $y \rightarrow 0$                | Asymptotic late branch with $V_g^{\text{F}} \rightarrow \beta_0$ .                        | Future limit                |

Table 1: Schematic chronology of the BPB/MBE biface branch. The pre-bounce and primordial-perturbation sectors remain outside the closed claims of Paper 0.

## 7 Effective potential and tachyonic-response threshold

The second derivative of the  $g$ -sector potential is

$$V_g^{\text{F}''}(y) = 6\beta_2 + 6\beta_3 y. \quad (59)$$

If  $\beta_2 > 0$  and  $\beta_3 < 0$ , the response changes sign at

$$y_{\text{crit}} = -\frac{\beta_2}{\beta_3}. \quad (60)$$

For the BPB/MBE branch lock,  $y_{\text{crit}} \simeq 0.600$ . The late-time trajectory gives  $y_0 \simeq 0.670$ , so that

$$y_0 > y_{\text{crit}}, \quad (61)$$

placing the present branch on the tachyonic-response side of  $V_g^{\text{F}}$ .

### 7.1 Meaning of “tachyonic”

The term tachyonic is used here in its standard field-theoretic sense: it denotes a negative curvature of the effective potential along the branch direction, not a superluminal propagation claim and not an ultraviolet catastrophe. The relevant response rate is controlled by the cosmological scale of the branch and must be compared with the Hubble rate. In this paper the tachyonic sign is used only for the effective dictionary: it explains why the growth descendant has the sign  $\beta_{\text{grow}} < 0$ .

### 7.2 Structural result

The sign of  $\beta_{\text{grow}}$  is therefore not chosen phenomenologically. Once the branch trajectory crosses  $y_{\text{crit}}$ , the HR potential curvature fixes the sign of the effective growth response. The magnitude is not fixed by the sign alone; it is set by the survival fraction and by the background growth correction. This separation is central:

$\text{sign from } V_g^{\text{F}''}(y_0) < 0, \quad \text{magnitude from } 1 - s_{\text{surv},0} \text{ and } G_{\text{corr}}.$

(62)

## 8 The survival descendant

### 8.1 Definition

The survival descendant is the effective fraction of the transition branch density that remains in the permanent observable component after the biface split has been encoded in the HR branch. In the branch-projection dictionary it is not defined as  $V_g^{\text{F}}(0)/V_g^{\text{F}}(y_t)$ . Instead,

$$s_{\text{surv,HR}} = \frac{\Pi_{\text{perm}}[U_{\text{HR}}]}{V_g^{\text{F}}(y_t)} = \frac{\beta_0 + \beta_1}{V_g^{\text{F}}(y_t)}. \quad (63)$$

The numerator is an action-level permanent projection. The denominator is the  $g$ -Friedmann transition branch density. This distinction is the core branch-projection result of the paper.

Since

$$\Pi_{\text{perm}}[U_{\text{HR}}] = \Omega_{\text{de}}, \quad V_g^{\text{F}}(y_t) = \Omega_{\text{de}} + \Delta\Omega, \quad (64)$$

Eq. (63) is equivalently

$$s_{\text{surv,HR}} = \frac{\Omega_{\text{de}}}{\Omega_{\text{de}} + \Delta\Omega}. \quad (65)$$

## 8.2 Rounded HR value and exact corpus lock

For the rounded HR branch lock,

$$\beta_0 + \beta_1 = 0.918 - 0.270 = 0.648, \quad V_g^F(y_t) = 0.810, \quad (66)$$

and therefore

$$s_{\text{surv,HR}} = \frac{0.648}{0.810} = 0.800. \quad (67)$$

The Friedmann source at  $y = 0$  is instead

$$V_g^F(0) = \beta_0 = 0.918, \quad (68)$$

so that

$$\frac{V_g^F(0)}{V_g^F(y_t)} = 1.133, \quad (69)$$

which is not a survival fraction. It is a diagnostic confirming that  $V_g^F(0)$ ,  $\Pi_{\text{perm}}[U_{\text{HR}}]$ , and  $V_g^F(y_t)$  must remain distinct throughout the paper.

The current numerical source lock used by the downstream corpus is

$$s_{\text{surv},0} = 0.7956924397708606, \quad 1 - s_{\text{surv},0} = 0.2043075602291394. \quad (70)$$

This value differs from the rounded HR algebra by about 0.54% in  $s_{\text{surv},0}$  and about 2.15% in the complementary fraction. In this paper, the rounded value explains the HR branch mechanism, while Eq. (70) is the source-locked numerical descendant used for acceleration-scale and application-paper calculations.

## 8.3 Non-circularity

Equation (63) is a theoretical dictionary relation. It does not use CMB lensing, redshift-space distortions, SPARC rotation curves, early-galaxy abundance, or neutron-star observables. Those sectors may later test the descendant, but they do not define the branch-projection rule. This is the first non-circularity requirement of Paper 0.

The comparison is summarized as

| Quantity                | Rounded HR branch value | Source-locked corpus value | Comment                   |
|-------------------------|-------------------------|----------------------------|---------------------------|
| $s_{\text{surv},0}$     | 0.800000                | 0.7956924397708606         | numerical lock            |
| $1 - s_{\text{surv},0}$ | 0.200000                | 0.2043075602291394         | acceleration-scale weight |

The small mismatch is not hidden. It is carried as a numerical source-lock correction, not as an additional fitted observational parameter.

## 9 Acceleration scale from the normalized HR $y$ -mode

The survival descendant fixes the permanent fraction of the transition branch density. The same branch algebra also fixes the complementary rounded HR fraction,

$$1 - s_{\text{surv,HR}} = \frac{\Delta\Omega}{V_g^F(y_t)} = 0.200. \quad (71)$$

The exact numerical corpus lock used for acceleration-scale applications is instead

$$1 - s_{\text{surv},0} = 0.2043075602291394. \quad (72)$$

This distinction is retained explicitly so that the analytic rounded branch algebra and the source-locked numerical application value are not confused.

The D2 source-lock result is that the local acceleration scale is not inserted as an independent phenomenological constant, but follows from the normalized HR  $y$ -mode at the lower Bianchi point and the effective de Sitter/Unruh conversion.

The purely algebraic branch identities are

$$V_g^{\text{F}'}(y_t) = 0, \quad (73)$$

$$\Delta\Omega = y_t^2 V_g^{\text{F}''}(y_t) = 0.162, \quad (74)$$

$$\frac{V_g^{\text{F}''}(y_t)}{V_g^{\text{F}}(y_t)} = \frac{20}{9}. \quad (75)$$

The mode mass in the HR-FLRW perturbation normalization used in the Comelli convention is therefore

$$\frac{M_y^2}{H_0^2} = \frac{V_g^{\text{F}''}(y_t)}{V_g^{\text{F}}(y_t)} = \frac{20}{9}. \quad (76)$$

Equivalently, the de Sitter index of the  $y$ -mode satisfies

$$\nu^2 = \frac{9}{4} - \frac{M_y^2}{H_0^2} = \frac{1}{36}, \quad \nu = \frac{1}{6}. \quad (77)$$

Using the Allen–Folacci de Sitter normalization [1] for this near-Higuchi scalar mode selects

$$m_{\text{FP}} = H_0 \sqrt{\frac{27}{8\pi^2 y_t^2 V_g^{\text{F}}(y_t)}} = 2.165824 H_0 \simeq 3.07 \times 10^{-33} \text{ eV}. \quad (78)$$

This is a closure-selected mass scale of the HR mode, not a new fitted acceleration parameter.

The corresponding rounded mode variance is

$$\langle \delta y^2 \rangle = y_t^2 = 0.090, \quad (79)$$

and the virial/equipartition fraction of the  $y$ -mode in the rounded branch vacuum is

$$\sin^2 \theta_y^{\text{HR}} = \frac{V_g^{\text{F}''}(y_t) \langle \delta y^2 \rangle}{V_g^{\text{F}}(y_t)} = \frac{1.800 \times 0.090}{0.810} = 0.200. \quad (80)$$

The source-locked application value replaces this rounded weight by Eq. (72). This is the numerical lock used by Paper VII and by the current corpus.

At the lower Bianchi point the reduced branch-flow interpretation gives  $H_f = 0$ . The  $f$ -sector de Sitter temperature is therefore zero, while the matter-coupled  $g$ -component sees the reduced temperature

$$T_{\text{eff}} = (1 - s_{\text{surv},0}) T_{\text{dS}}^{(g)}. \quad (81)$$

Equating this temperature to the local Unruh temperature gives

$$\frac{\hbar g_N(r_t)}{2\pi c k_B} = (1 - s_{\text{surv},0}) \frac{\hbar H_0}{2\pi k_B}, \quad (82)$$

and hence

$$g_N(r_t) = c H_0 (1 - s_{\text{surv},0}) \equiv a_0^{\text{BPB}}. \quad (83)$$

With the primary numerical locks

$$H_0 = 68.910590682276 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad s_{\text{surv},0} = 0.7956924397708606, \quad (84)$$

one obtains

$$a_0^{\text{BPB}} = 1.367856592595 \times 10^{-10} \text{ m s}^{-2}. \quad (85)$$

The corresponding de Sitter temperatures are

$$T_{\text{dS}} = 2.714866240842 \times 10^{-30} \text{ K}, \quad T_{\text{eff}} = 5.546676980150 \times 10^{-31} \text{ K}. \quad (86)$$

This D2 closure is local to the acceleration-scale descendant and is internally source-locked under the documented H1–H4 convention bundle. It does not by itself derive the full response hierarchy of Papers I–VII, and it is not an unconditional action-level derivation of a MOND interpolation function. The response functions entering growth, Weyl auto/cross projections, galaxy-scale kernels, and neutron-star tidal response remain separate descendants or companion-paper calculations.

## 10 The growth-drag descendant

### 10.1 Definition and physical distinction

The coefficient  $\beta_{\text{grow}}$  is an effective growth descendant. It must not be interpreted as a static fifth force, a direct modification of local orbital gravity, or a universal replacement  $G_N \rightarrow G_{\text{eff}}$  in all environments. In the BPB/MBE dictionary it acts on cosmological growth through the cold-sector split and through the background growth history.

The minimal dictionary relation is

$$\beta_{\text{grow}} \simeq -(1 - s_{\text{surv},0}) G_{\text{corr}}, \quad (87)$$

where  $G_{\text{corr}}$  is the growth-history correction obtained by integrating the background growth equation with the BPB/MBE branch expansion history. The minus sign is assigned by the negative-curvature side of the HR branch identified in Section 7.

### 10.2 Background growth correction

A convenient definition is

$$G_{\text{corr}} = \frac{D_{\text{BPB}}(a=1)}{D_{\text{ref}}(a=1)}, \quad (88)$$

where  $D(a)$  is the linear growth solution normalized on the same early-time convention and “ref” denotes the chosen  $\Lambda$ CDM reference background. The branch lock gives

$$G_{\text{corr}} \simeq 1.0337. \quad (89)$$

Combining this value with  $s_{\text{surv},0}^{\text{HR}} = 0.8000$  gives

$$\beta_{\text{grow}}^{\text{HR}} \simeq -(0.2000)(1.0337) \simeq -0.2067. \quad (90)$$

Using instead the exact source lock  $s_{\text{surv},0} = 0.7956924397708606$  gives  $\beta_{\text{grow}} \simeq -0.2112$ , close to the current effective/MAP comparison  $\beta_{\text{grow}} \simeq -0.2107$ .

### 10.3 Numerical comparison

| Formula                    | Dictionary value | Effective/MAP comparison | Offset         |
|----------------------------|------------------|--------------------------|----------------|
| $-(1 - 0.8000) 1.0337$     | $-0.2067$        | $-0.2107$                | $\simeq 1.9\%$ |
| $-(1 - 0.79569244) 1.0337$ | $-0.2112$        | $-0.2107$                | $\simeq 0.2\%$ |

## 10.4 Interpretation

The sign comes from the HR branch curvature. The magnitude comes from the split-relic survival deficit and from the background growth correction. This is the second non-circularity requirement of Paper 0:  $\beta_{\text{grow}}$  is predicted by the same branch dictionary that predicts  $s_{\text{surv},0}$ . What is not claimed here is a completed first-principles derivation of the full gauge-invariant HR perturbation sector in all regimes; that belongs to the EFT closure layer.

## 11 Perturbation dictionary and EFT closure

### 11.1 Exact HR response versus effective closure

The HR theory has a full perturbation system involving both metrics. The BPB/MBE phenomenological implementation uses an effective dictionary that projects the HR response into window-averaged functions such as  $\mu_{\text{eff}}(a)$  and  $\Sigma_{\text{eff}}(a)$ . It is essential to distinguish these levels:

- (i) the HR perturbation equations are the underlying theory;
- (ii) the BPB/MBE dictionary identifies the branch-generated descendants;
- (iii) the EFT closure maps those descendants onto observable windows.

The present paper derives the first two levels and states the closure logic for the third. It does not claim a final all-scale EFT proof.

### 11.2 Quasi-static schematic form

In a quasi-static regime, one may write schematically

$$\mu(k, a) = \frac{k^2 - m^2 a^2 \Sigma_1(a)}{k^2 - m^2 a^2 \Sigma_2(a)}, \quad (91)$$

where the  $\Sigma_i$  are branch-dependent combinations of HR potential derivatives and background quantities. Equation (91) is a schematic dictionary expression, not the final all-gauge perturbation theory. Its purpose is to show why the small-scale limit recovers GR while the large-window response can inherit the sign of the branch curvature:

$$\mu(k \gg am, a) \rightarrow 1, \quad \mu(k \lesssim am, a) \rightarrow \text{branch-dependent response}. \quad (92)$$

### 11.3 EFT closure used in applications

The effective growth closure may be written as

$$\mu_{\text{BPB}}(a) = 1 + \beta_{\text{grow}} S_p(a), \quad (93)$$

where  $S_p(a)$  is a window function tied to the branch activation. Lensing closures may require a separate Weyl response, for instance

$$\Sigma_{\text{BPB}}(a) = 1 + c_\Sigma u(a), \quad (94)$$

with  $u(a)$  a branch-memory function. Paper 0 does not impose a universal equality  $\mu = \Sigma$ . Instead it defines which quantities are HR descendants and which are window-dependent closures.

## 11.4 Guardrails

The safe claims at this stage are:

- the HR action is ghost-free at the level established by the HR construction;
- the late-time branch has  $V_g^{\text{F}''}(y_0) < 0$ , fixing the sign of the growth descendant;
- the high- $k$  linear limit can recover GR in the effective response;
- nonlinear screening, full gradient stability, and a complete gauge-invariant perturbation analysis remain outside the scope of Paper 0.

## 12 Parameter counting and information criteria

Information criteria count fitted degrees of freedom, not quantities predicted after the fit. We use

$$\text{AIC} = \chi_{\min}^2 + 2k, \quad \text{BIC} = \chi_{\min}^2 + k \ln N, \quad (95)$$

where  $k$  is the number of fitted parameters and  $N$  is the number of data points.

### 12.1 The counting rule

BPB/MBE is not a zero-parameter extension of  $\Lambda$ CDM. Rather, it is a same-order-parameter late-acceleration-sector replacement in which a structural split  $\Delta\Omega$  generates the effective descendants  $s_{\text{surv},0}$  and  $\beta_{\text{grow}}$  through the HR dictionary. A representative comparison is therefore

| Model         | Representative fitted parameters                                                     | Count |
|---------------|--------------------------------------------------------------------------------------|-------|
| $\Lambda$ CDM | $H_0, \Omega_b, \Omega_c, \Omega_\Lambda, A_s, n_s, \tau$                            | 7     |
| BPB/MBE       | $H_0, \Omega_b, \Omega_{m,\text{late}}, \Delta\Omega, a_{\text{bg}}, A_s, n_s, \tau$ | 8     |

The exact absolute value of  $k$  depends on the comparator convention. For example, the usual flat CMB  $\Lambda$ CDM parameterization is often counted with six parameters, while a background-plus-growth comparison may expose  $H_0$  and density components differently. The important model-selection rule is independent of that convention:

$$s_{\text{surv},0} = f_{\text{HR}}(\Delta\Omega, \dots), \quad \beta_{\text{grow}} = g_{\text{HR}}(\Delta\Omega, a_{\text{bg}}, \dots), \quad (96)$$

so they are not counted as independent parameters when the HR dictionary is enforced.

### 12.2 Conservative wording

The correct statement is not that BPB/MBE is parameter-free. The correct statement is:

BPB/MBE is a same-order-parameter replacement of the late-acceleration sector in which a structural split generates multiple effective descendants. Information criteria should penalize the structural split and branch location, not each cross-predicted descendant as though it were independently fitted.

This is the parameter-counting logic to be used in subsequent observational papers.



### 13 Effective late-acceleration equation of state

The BPB/MBE branch does not need to introduce a fitted constant vacuum fluid in order to describe late acceleration. The relevant object in Paper 0 is the HR Friedmann-source polynomial  $V_g^F(y)$  evaluated along the branch. One may therefore define an effective acceleration equation-of-state diagnostic by asking how this source changes with redshift.

Using the local branch relation

$$y(z) = y_t(1 + z), \quad y_t = 0.300, \quad (97)$$

and treating  $V_g^F[y(z)]$  as the effective acceleration density, the usual continuity-form definition gives

$$w_{\text{eff}}(z) = -1 + \frac{1+z}{3} \frac{d \ln V_g^F[y(z)]}{dz} = -1 + \frac{y_t(1+z)}{3} \frac{V_g^{F'}[y_t(1+z)]}{V_g^F[y_t(1+z)]}. \quad (98)$$

This is a diagnostic of the HR branch, not a new fitted fluid.

Since the present point is a Bianchi stationary point,

$$V_g^{F'}(y_t) = 0, \quad (99)$$

Eq. (98) gives the exact present value

$$w_{\text{eff}}(0) = -1. \quad (100)$$

The local slope in the usual  $w_0$ - $w_a$  convention is small and positive,

$$w_a \simeq +0.0667. \quad (101)$$

Because the same Bianchi polynomial has a second root at  $y = 0.900 = 3y_t$ , the branch predicts a second exact crossing of  $w_{\text{eff}} = -1$  at

$$z = \frac{0.900}{0.300} - 1 = 2. \quad (102)$$

Between the two crossings the branch forms a shallow hill: over  $0 \leq z \leq 2$ , the maximum value is approximately

$$w_{\text{eff}}^{\text{max}} \simeq -0.935 \quad \text{near} \quad z \simeq 1.18. \quad (103)$$

The expression remains in the positive-source branch until the first zero of  $V_g^F$ ,

$$y \simeq 1.647, \quad z \simeq 4.49. \quad (104)$$

This is a direct falsification channel. BPB/MBE predicts a specific non-constant late-acceleration diagnostic:  $w_0 = -1$  exactly,  $w_a > 0$  locally, and a second crossing at  $z = 2$ . The point is not that the model follows any particular empirical  $w_0$ - $w_a$  preference. The point is stronger: the HR branch fixes a distinct shape that can be confronted with distance data.

| Quantity                                           | Value           |
|----------------------------------------------------|-----------------|
| $y_t$                                              | 0.300           |
| $w_{\text{eff}}(0)$                                | -1 exact        |
| $w_a$                                              | +0.0667         |
| second $w = -1$ crossing                           | $z = 2.000$     |
| $w_{\text{eff}}^{\text{max}}$ on $0 \leq z \leq 2$ | $\simeq -0.935$ |
| redshift of maximum                                | $\simeq 1.18$   |
| positive-source validity                           | $z < 4.49$      |

Table 2: Parameter-free late-acceleration equation-of-state diagnostic implied by the BPB/MBE branch.

## 14 Applications and falsification channels

The present work is not an all-probe fit. Nevertheless, the HR dictionary is useful only if it opens falsification channels. We therefore list the main channels as downstream projections of the dictionary, not as inputs to it. The companion manuscripts are the natural place where these projections are tested.

| Paper | Channel                                     | What would falsify or stress the descendant                                                                                        | Paper 0 status |
|-------|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|----------------|
| I     | Low-redshift distances, growth, and lensing | Failure of the shared $s_{\text{surv},0}$ , $\beta_{\text{grow}}$ , and Weyl/growth split under a common parameter-counting rule   | Application    |
| II    | SPARC and galaxy amplitudes                 | Wrong sign or failed raw rotation-curve amplitude response after source-only kernel injection                                      | Application    |
| III   | CMB/lensing response                        | Failure of the survival/face-minus CMB mechanism or Weyl-sector response under spectra/likelihood checks                           | Application    |
| IV    | Neutron stars                               | Loss of EOS-coherent strict-dictionary tidal response or topology consistency                                                      | Application    |
| V     | Early galaxies                              | Inability of the survival-normalized mapper to reproduce the high-redshift abundance structure under stated priors                 | Application    |
| VI    | $E_G$ / Weyl-matter kernel                  | Failure of the projected $r_{W\delta}$ closure in the appropriate auto/cross amplitude space                                       | Application    |
| VII   | BTFR / $a_0$                                | Failure of $a_0 = cH_0(1 - s_{\text{surv},0})$ as a fixed normalization and raw baryonic interpolation scale                       | Application    |
| –     | Late-acceleration $w_{\text{eff}}(z)$       | Failure of the branch-shape prediction $w_0 = -1$ , Paper 0 pre- $w_a \simeq 0.0667$ , and the second $w = -1$ crossing at $z = 2$ | diction        |

These channels are valuable because they test descendants in independent regimes. They are deliberately not used to define the branch in this paper. In particular, SPARC, neutron-star, early-galaxy, BTFR,  $E_G$ , and CMB likelihood results should not be used in Paper 0 as foundations of the HR dictionary. They belong to the companion papers and to the corpus-level source-lock manifest.

## 15 Scope of this paper

### 15.1 What is claimed

This paper claims:

- (1) a non-circular  $\text{HR} \rightarrow \text{BPB/MBE}$  effective dictionary;
- (2) the identification of the biface-genesis branch as the HR trajectory generating the dictionary;
- (3) the branch-projection derivation of the survival descendant  $s_{\text{surv},0} = (\beta_0 + \beta_1)/V_g^F(y_t)$ ;
- (4) the D2 acceleration-scale descendant  $a_0^{\text{BPB}} = cH_0(1 - s_{\text{surv},0})$  from the normalized HR  $y$ -mode, including the closure-selected value  $m_{\text{FP}} = 2.165824 H_0$  and the exact numerical lock  $a_0^{\text{BPB}} = 1.367856592595 \times 10^{-10} \text{ m s}^{-2}$ ;
- (5) the derivation of the growth-drag descendant  $\beta_{\text{grow}} \simeq -(1 - s_{\text{surv},0})G_{\text{corr}}$  at the effective-dictionary level;
- (6) the structural origin of the sign  $\beta_{\text{grow}} < 0$  from  $V_g^{F''}(y_0) < 0$ ;
- (7) the dynamical interpretation of the present lower Bianchi point  $y_t = 0.300$  as the reduced branch-flow stationary endpoint;

- (8) the effective late-acceleration diagnostic  $w_{\text{eff}}(z)$  derived from  $V_g^{\text{F}}[y(z)]$ , including  $w_{\text{eff}}(0) = -1$  and the second crossing at  $z = 2$ ;
- (9) the parameter-counting rule that treats  $s_{\text{surv},0}$  and  $\beta_{\text{grow}}$  as outputs, not independent inputs;
- (10) the separation between the action-level HR potential, the Friedmann-source polynomial, and the effective observable projection.

## 15.2 What is not claimed

This paper does not claim:

- (1) a full Planck MCMC analysis;
- (2) a complete HR perturbative stability proof for all modes and all gauges;
- (3) a nonlinear Vainshtein screening proof;
- (4) a final EFT projection from  $U_{\text{HR}}(X, y)$  and  $V_g^{\text{F}}(y)$  to every observable window;
- (5) proof of pre-bounce fossil compact objects;
- (6) a replacement of the Kerr exterior for Sgr A\* or for standard orbital tests;
- (7) that Paper 0 itself is the final SPARC, BTFR, neutron-star, KiDS, ACT, CMB,  $E_G$ , or early-galaxy likelihood paper;
- (8) a completed primordial perturbation spectrum from the  $1.647 > y > 1$  phase;
- (9) a source-matched  $f$ -sector closure fixing  $\beta_4$ ;
- (10) an unconditional action-level derivation of a MOND interpolation function from D2 alone;
- (11) that  $\Lambda\text{CDM}$  is ruled out.

The limitations are therefore part of the scope, not a retreat from the central result. Paper 0 defines the theoretical dictionary. The companion papers test it, but Paper 0 does not import those tests as assumptions.

## 15.3 Corpus-release claim boundary

In a simultaneous release, the phrase “BPB/MBE corpus” refers to the union of Paper 0 and the companion manuscripts. The phrase “Paper 0 claim” refers only to the HR dictionary and its source-locked descendants. This distinction is used throughout the release: the corpus may contain observational passes, raw rotation-curve gates, CMB mechanism gates, or compact-object audits, while Paper 0 itself remains the derivation and bookkeeping layer.

## 16 Conclusion

We have formulated BPB/MBE as an effective dictionary generated by a late-time branch of ghost-free Hassan–Rosen bigravity. The key object is the biface-genesis branch, the HR cosmological trajectory for which the scale-factor ratio  $y(a)$  crosses the potential-curvature threshold  $y_{\text{crit}} = -\beta_2/\beta_3$  and reaches  $y_0 > y_{\text{crit}}$  today. This crossing places the present branch on the negative-curvature side of the  $g$ -Friedmann source polynomial and fixes the sign of the effective growth descendant.

The v0.5 formulation adds a dynamical reading of the same lock. The present value  $y_t = 0.300$  is the lower Bianchi stationary point of the reduced branch flow. Evaluating the transition Friedmann source and the permanent action projection at this point gives  $V_g^F(y_t) = 0.810$ ,  $U_{\text{HR}}(1, 0) = 0.648$ , and therefore  $\Delta\Omega = 0.162$ .

The branch-projection formulation clarifies the origin of the survival factor. The permanent component is not the raw Friedmann-source value  $V_g^F(0)$ . It is selected from the action-level HR interaction by the branch-projection rule

$$\Omega_{\text{de}} = \Pi_{\text{perm}}[U_{\text{HR}}] = U_{\text{HR}}(1, 0) = \beta_0 + \beta_1. \quad (105)$$

The rounded HR survival value is therefore generated by the permanent-to-transition ratio

$$s_{\text{surv,HR}} = \frac{\beta_0 + \beta_1}{V_g^F(y_t)} = \frac{0.648}{0.810} = 0.800, \quad (106)$$

not by  $V_g^F(0)/V_g^F(y_t)$ . The latter ratio is larger than unity in the branch lock and is used only as a consistency guardrail. The exact numerical corpus lock is  $s_{\text{surv},0} = 0.7956924397708606$ .

The growth drag is generated, at the effective-dictionary level, by

$$\beta_{\text{grow}} \simeq -(1 - s_{\text{surv},0})G_{\text{corr}}. \quad (107)$$

For the rounded HR branch values, these give  $s_{\text{surv,HR}} = 0.800$  and  $\beta_{\text{grow}}^{\text{HR}} \simeq -0.2067$ . With the exact corpus lock, one obtains  $\beta_{\text{grow}} \simeq -0.2112$ , close to the independently inferred effective value  $\beta_{\text{grow}} \simeq -0.2107$ . The sign comes from the negative-curvature side of the HR branch; the magnitude comes from the split-relic survival deficit and the background growth correction.

The same formulation fixes the status of the  $f$  sector. The condition  $V_f^F(y_t) = 0$  is not imposed in the minimal Paper 0 dictionary, because with the current  $g$ -sector branch it would force  $\beta_4 = 0$ . Instead  $\beta_4$  is kept as a residual branch-closure parameter.

The normalized HR  $y$ -mode at the lower Bianchi point also closes the acceleration-scale descendant. The D2 source-lock gives  $\nu = 1/6$ , selects  $m_{\text{FP}} = 2.165824 H_0$ , and yields  $a_0^{\text{BPB}} = cH_0(1 - s_{\text{surv},0}) = 1.367856592595 \times 10^{-10} \text{ m s}^{-2}$  through Unruh-de Sitter balance for the primary exact corpus locks. This D2 statement is source-locked under the documented H1–H4 convention bundle; it is not, by itself, a full nonrelativistic MOND action-level derivation.

The same branch also predicts a parameter-free late-acceleration diagnostic. With  $y(z) = y_t(1 + z)$ , the effective equation of state is  $w_{\text{eff}}(z) = -1 + y_t(1 + z)V_g^{F'}[y_t(1 + z)]/[3V_g^F(y_t(1 + z))]$ . Hence  $w_{\text{eff}}(0) = -1$  exactly, the local slope is positive,  $w_a \simeq 0.0667$ , and the second Bianchi point gives a second  $w = -1$  crossing at  $z = 2$ .

The main consequence for model comparison is that BPB/MBE should not be counted as a model where  $s_{\text{surv},0}$  and  $\beta_{\text{grow}}$  are freely adjusted in every observational sector. When the HR dictionary is enforced, they are outputs of the structural branch. BPB/MBE is therefore not a zero-parameter extension of  $\Lambda\text{CDM}$ , but a same-order-parameter replacement of the late-acceleration sector in which a structural split generates cross-sector descendants.

This establishes the purpose of Paper 0: to define the HR origin of the BPB/MBE dictionary, to state its parameter-counting rule, and to separate derived theoretical structure from downstream observational applications. In the HAL corpus release, the companion manuscripts are therefore not decorative appendices and not hidden assumptions. They are the sector-by-sector tests of the descendants derived here. The release root provides the navigation layer: `00_README_CORPUS` for human orientation, `00_MANIFEST_CORPUS` for file provenance, `00_CLAIMS_INDEX` for source-locked claims, and `00_REVIEWER_ROUTE` for a skeptical audit path. Paper 0 is the theoretical entry point; the corpus is the falsification map.

## A HR action-level potential and Friedmann-source polynomials

For the FLRW square-root eigenvalues

$$S^\mu{}_\nu = \text{diag}(X, y, y, y), \quad (108)$$

the elementary symmetric polynomials give

$$e_0 = 1, \quad (109)$$

$$e_1 = X + 3y, \quad (110)$$

$$e_2 = 3Xy + 3y^2, \quad (111)$$

$$e_3 = 3Xy^2 + y^3, \quad (112)$$

$$e_4 = Xy^3. \quad (113)$$

Thus the action-level interaction polynomial is

$$\begin{aligned} U_{\text{HR}}(X, y) &= \beta_0 + \beta_1(X + 3y) + \beta_2(3Xy + 3y^2) \\ &\quad + \beta_3(3Xy^2 + y^3) + \beta_4Xy^3. \end{aligned} \quad (114)$$

The permanent branch projection used in Paper 0 is

$$\Pi_{\text{perm}}[U_{\text{HR}}] = U_{\text{HR}}(1, 0) = \beta_0 + \beta_1. \quad (115)$$

Varying the action with respect to the  $g$ -metric lapse gives the  $g$ -Friedmann source polynomial

$$V_g^{\text{F}}(y) = \beta_0 + 3\beta_1y + 3\beta_2y^2 + \beta_3y^3. \quad (116)$$

Varying with respect to the  $f$ -metric lapse gives the corresponding  $f$ -sector source polynomial

$$V_f^{\text{F}}(y) = \beta_4y^3 + 3\beta_3y^2 + 3\beta_2y + \beta_1. \quad (117)$$

The schematic unprojected Friedmann system is

$$3H_g^2 = \rho_m/M_g^2 + m^2V_g^{\text{F}}(y), \quad (118)$$

$$3H_f^2 = m^2V_f^{\text{F}}(y)/y^3. \quad (119)$$

The important convention is therefore:

$$\Omega_{\text{de}} = U_{\text{HR}}(1, 0) = \beta_0 + \beta_1, \quad V_g^{\text{F}}(0) = \beta_0. \quad (120)$$

These are different HR projections. Their distinction is what makes the survival descendant well-defined.

## B Resolution of the branch locks

The minimal locks are

$$\beta_0 + \beta_1 = \Omega_{\text{de}}, \quad (121)$$

$$V_g^{\text{F}}(y_t) = \Omega_{\text{de}} + \Delta\Omega, \quad (122)$$

$$V_g^{\text{F}'}(y_t) = 0, \quad (123)$$

$$y_{\text{crit}} = -\beta_2/\beta_3. \quad (124)$$

With the branch normalization

$$y_t = 0.300, \quad y_{\text{crit}} = 0.600, \quad \beta_3 = -1, \quad (125)$$

Eq. (124) gives

$$\beta_2 = 0.600. \quad (126)$$

The stationary condition gives

$$3\beta_1 + 6\beta_2 y_t + 3\beta_3 y_t^2 = 0 \quad \Rightarrow \quad \beta_1 = -0.270. \quad (127)$$

The permanent projection lock then gives

$$\beta_0 = \Omega_{\text{de}} - \beta_1 = 0.648 + 0.270 = 0.918. \quad (128)$$

Evaluating the  $g$ -Friedmann source at  $y_t$  gives

$$V_g^{\text{F}}(y_t) = 0.810, \quad (129)$$

and therefore

$$\Delta\Omega = V_g^{\text{F}}(y_t) - \Omega_{\text{de}} = 0.162. \quad (130)$$

Under the same branch normalization,

$$\Delta\Omega = y_t^2(3 - 4y_t) = 0.162, \quad (131)$$

which makes the split a branch-geometry quantity in this lock. The raw value

$$V_g^{\text{F}}(0) = \beta_0 = 0.918 \quad (132)$$

is explicitly not the permanent component.

## C Potential curvature and critical value

From Eq. (19),

$$V_g^{\text{F}''}(y) = 6\beta_2 + 6\beta_3 y. \quad (133)$$

The curvature changes sign at

$$y_{\text{crit}} = -\frac{\beta_2}{\beta_3}. \quad (134)$$

For the lock values  $\beta_2 \simeq 0.600$  and  $\beta_3 \simeq -1.000$ , this gives

$$y_{\text{crit}} \simeq 0.600. \quad (135)$$

The branch is tachyonic-response when  $y > y_{\text{crit}}$ . The current dictionary target has  $y_0 \simeq 0.670$ , so  $V_g^{\text{F}''}(y_0) < 0$ .

## D Derivation of the survival descendant

The rounded HR survival descendant is defined by the ratio of the permanent action-level projection to the transition Friedmann-source density:

$$s_{\text{surv,HR}} = \frac{\Pi_{\text{perm}}[U_{\text{HR}}]}{V_g^{\text{F}}(y_t)} = \frac{U_{\text{HR}}(1, 0)}{V_g^{\text{F}}(y_t)} = \frac{\beta_0 + \beta_1}{V_g^{\text{F}}(y_t)}. \quad (136)$$

Using

$$\beta_0 + \beta_1 = 0.648, \quad V_g^{\text{F}}(y_t) = 0.810, \quad (137)$$

one obtains

$$s_{\text{surv,HR}} = \frac{0.648}{0.810} = 0.800. \quad (138)$$

The exact corpus value is then recorded as the numerical source lock

$$s_{\text{surv},0} = 0.7956924397708606, \quad 1 - s_{\text{surv},0} = 0.2043075602291394. \quad (139)$$

This exact value is used for numerical acceleration-scale propagation and for application-paper comparisons.

The consistency guardrail is

$$\frac{V_g^{\text{F}}(0)}{V_g^{\text{F}}(y_t)} = \frac{\beta_0}{V_g^{\text{F}}(y_t)} = \frac{0.918}{0.810} = 1.133 > 1, \quad (140)$$

which is not a survival fraction. This confirms that  $V_g^{\text{F}}(0)$ ,  $U_{\text{HR}}(1,0)$ , and  $V_g^{\text{F}}(y_t)$  must be kept distinct. This calculation does not use perturbative, galactic, CMB, or compact-object data.

## E Derivation of the growth-drag descendant

The background growth equation can be written in the form

$$D'' + \left[ 2 + \frac{d \ln H}{d \ln a} \right] D' = \frac{3}{2} \frac{\Omega_{m,\text{eff}}(a)}{E(a)^2} D, \quad (141)$$

where primes denote derivatives with respect to  $\ln a$  and  $E(a) = H(a)/H_0$ . The correction factor is

$$G_{\text{corr}} = \frac{D_{\text{BPB}}(1)}{D_{\text{ref}}(1)}. \quad (142)$$

The minimal growth-drag descendant is then

$$\beta_{\text{grow}} \simeq -(1 - s_{\text{surv},0}) G_{\text{corr}}. \quad (143)$$

For  $G_{\text{corr}} \simeq 1.0337$  and the rounded HR value  $s_{\text{surv,HR}} = 0.8000$ ,

$$\beta_{\text{grow}} \simeq -0.2067. \quad (144)$$

For the exact source lock  $s_{\text{surv},0} = 0.7956924397708606$ ,

$$\beta_{\text{grow}} \simeq -0.2112. \quad (145)$$

These values bracket the effective/MAP comparison  $\beta_{\text{grow}} \simeq -0.2107$ , with the exact source-locked value giving the closest numerical match.

## F AIC/BIC counting

The information criteria are

$$\text{AIC} = \chi_{\text{min}}^2 + 2k, \quad \text{BIC} = \chi_{\text{min}}^2 + k \ln N. \quad (146)$$

The relevant question is not how many observable consequences a model has, but how many independent degrees of freedom were fitted to obtain them. In the HR dictionary,

$$s_{\text{surv},0} = f_{\text{HR}}(\Delta\Omega, \dots), \quad \beta_{\text{grow}} = g_{\text{HR}}(\Delta\Omega, a_{\text{bg}}, \dots). \quad (147)$$

Therefore a model comparison that enforces the dictionary should not add separate penalties for  $s_{\text{surv},0}$  and  $\beta_{\text{grow}}$  as if they were free nuisance parameters. A conservative minimal count is  $k_{\text{BPB}} = 8$  versus  $k_{\Lambda\text{CDM}} = 7$ , giving an AIC penalty difference of +2 before any improvement in  $\chi^2$  is considered.

## G Consistency ledger

The following entries summarize the algebraic consistency ledger of the branch-projection dictionary. They are retained here to make the numerical definitions unambiguous and reproducible.

- $U_{\text{HR}}(X, y)$  is the action-level HR interaction potential, defined in Eq. (13).
- $V_g^{\text{F}}(y)$  is the  $g$ -Friedmann source polynomial, defined in Eq. (19).
- $U_{\text{HR}}(1, 0) = \beta_0 + \beta_1 = 0.648$  is the permanent branch projection and is identified with  $\Omega_{\text{de}}$ .
- $V_g^{\text{F}}(0) = \beta_0 \simeq 0.918$  is the raw Friedmann-source value at  $y = 0$  and is not  $\Omega_{\text{de}}$ .
- $V_g^{\text{F}}(y_t) = 0.810$  is the transition branch density.
- $\Delta\Omega = 0.162$  is both  $V_g^{\text{F}}(y_t) - U_{\text{HR}}(1, 0)$  and, in the branch normalization used here,  $y_t^2(3 - 4y_t)$ .
- The Bianchi polynomial  $\mathcal{B}(y) = \beta_1 + 2\beta_2 y + \beta_3 y^2$  factorizes as  $-(y - 0.300)(y - 0.900)$  for the lock values.
- $V_g^{\text{F}}(0)/V_g^{\text{F}}(y_t) = 1.133$  is a consistency guardrail only; it is not a survival factor.
- $s_{\text{surv,HR}} = 0.800$  is the rounded branch-projection ratio  $(\beta_0 + \beta_1)/V_g^{\text{F}}(y_t)$ .
- $s_{\text{surv},0} = 0.7956924397708606$  and  $1 - s_{\text{surv},0} = 0.2043075602291394$  are the exact corpus numerical source locks.
- $a_0^{\text{BPB}} = cH_0(1 - s_{\text{surv},0}) = 1.367856592595 \times 10^{-10} \text{ m s}^{-2}$  is the D2 acceleration-scale descendant from the normalized HR  $y$ -mode and exact corpus survival lock.
- $m_{\text{FP}} = 2.165824 H_0 \simeq 3.0711 \times 10^{-33} \text{ eV}$  is the closure-selected mass scale of the same mode.
- $y_{\text{crit}} = 0.600$  is  $-\beta_2/\beta_3$ .
- $y_0 \simeq 0.670$  is the present branch value from the HR/Bianchi trajectory.
- $\beta_{\text{grow}}^{\text{HR}} \simeq -0.2067$  follows from the rounded survival value and  $-(1 - s_{\text{surv,HR}})G_{\text{corr}}$ .
- $\beta_{\text{grow}} \simeq -0.2112$  follows from the exact source lock;  $\beta_{\text{grow}}^{\text{eff}} \simeq -0.2107$  is the effective/MAP comparison.
- $V_f^{\text{F}}(y_t) = 0$  is not imposed, since it would force  $\beta_4 = 0$  for the current  $g$ -sector branch.
- $\beta_4$  remains a residual  $f$ -sector and EFT closure parameter.
- The late-acceleration diagnostic predicts  $w_{\text{eff}}(0) = -1$ ,  $w_a \simeq +0.0667$ , and a second  $w = -1$  crossing at  $z = 2$ .



The minimal consistency assertions are:

$$0 < s_{\text{surv},0} < 1, \quad (148)$$

$$s_{\text{surv},\text{HR}} = \frac{\beta_0 + \beta_1}{V_g^{\text{F}}(y_t)}, \quad (149)$$

$$a_0^{\text{BPB}} = cH_0(1 - s_{\text{surv},0}) \quad \text{with the exact corpus source lock,} \quad (150)$$

$$V_g^{\text{F}}(y_t) - (\beta_0 + \beta_1) = \Delta\Omega, \quad (151)$$

$$\frac{V_g^{\text{F}}(0)}{V_g^{\text{F}}(y_t)} > 1 \quad \text{and is not a survival factor,} \quad (152)$$

$$y_0 > y_{\text{crit}}, \quad (153)$$

$$\beta_{\text{grow}} = -(1 - s_{\text{surv},0})G_{\text{corr}}, \quad (154)$$

$$V_f^{\text{F}}(y_t) = 0 \quad \text{is not a minimal lock.} \quad (155)$$

## H Dynamic lock algebra

For the lock values

$$\beta_1 = -0.270, \quad \beta_2 = 0.600, \quad \beta_3 = -1, \quad (156)$$

the Bianchi stationary polynomial is

$$\mathcal{B}(y) = \beta_1 + 2\beta_2 y + \beta_3 y^2 = -0.270 + 1.200y - y^2. \quad (157)$$

It factorizes exactly as

$$\mathcal{B}(y) = -(y - 0.300)(y - 0.900). \quad (158)$$

Thus the branch contains two stationary Bianchi points:

$$y_{B1} = 0.300, \quad y_{B2} = 0.900. \quad (159)$$

The lower point is the present branch value. At this point

$$V_g^{\text{F}}(0.300) = 0.810, \quad (160)$$

$$U_{\text{HR}}(1, 0) = \beta_0 + \beta_1 = 0.648, \quad (161)$$

$$\Delta\Omega = 0.810 - 0.648 = 0.162. \quad (162)$$

The same result can be written, in the present branch normalization, as

$$\Delta\Omega = y_t^2(3 - 4y_t) = 0.300^2(3 - 1.200) = 0.162. \quad (163)$$

## I Late-acceleration diagnostic values

The effective equation-of-state diagnostic follows from

$$w_{\text{eff}}(z) = -1 + \frac{y_t(1+z)}{3} \frac{V_g^{\text{F}'}[y_t(1+z)]}{V_g^{\text{F}}[y_t(1+z)]}. \quad (164)$$

Using  $y_t = 0.300$  and the lock polynomial, the present derivative vanishes because  $V_g^{\text{F}'}(0.300) = 0$ . The second derivative root occurs at  $y = 0.900$ , giving the second crossing at  $z = 2$ . The source remains positive until the first positive zero of  $V_g^{\text{F}}$ ,  $y \simeq 1.647$ , corresponding to  $z \simeq 4.49$ .

| Diagnostic                                     | Value                                |
|------------------------------------------------|--------------------------------------|
| $w_{\text{eff}}(0)$                            | $-1$ exact                           |
| $w_a$                                          | $+0.0667$                            |
| second crossing                                | $z = 2.000$                          |
| maximum on $0 \leq z \leq 2$                   | $w \simeq -0.935$ at $z \simeq 1.18$ |
| validity before $V_g^{\text{F}} \rightarrow 0$ | $z < 4.49$                           |

## J D2 acceleration-scale source lock

This appendix records the v0.7 source lock for the D2 acceleration-scale closure. The audit concerns the descendant

$$a_0^{\text{BPB}} = cH_0(1 - s_{\text{surv},0}), \quad (165)$$

not the full response hierarchy and not an unconditional action-level derivation of a MOND interpolation function.

| Quantity                                     | Value                                                | Status                      |
|----------------------------------------------|------------------------------------------------------|-----------------------------|
| $V_g^{\text{F}'}(y_t)$                       | 0                                                    | Bianchi point               |
| $\Delta\Omega = y_t^2 V_g^{\text{F}''}(y_t)$ | 0.162                                                | exact branch identity       |
| $V_g^{\text{F}''}(y_t)/V_g^{\text{F}}(y_t)$  | 20/9                                                 | exact                       |
| $\nu^2$                                      | 1/36                                                 | exact                       |
| $\nu$                                        | 1/6                                                  | exact                       |
| $f(\nu)$                                     | 27/2                                                 | Allen–Folacci normalization |
| $m_{\text{FP}}/H_0$                          | 2.165824                                             | closure-selected            |
| $m_{\text{FP}}$                              | $3.07 \times 10^{-33}$ eV                            | cosmological scale          |
| $\langle \delta y^2 \rangle$                 | 0.090                                                | normalized $y$ -mode        |
| $\sin^2 \theta_y^{\text{HR}}$                | 0.200                                                | rounded HR mode fraction    |
| $1 - s_{\text{surv},0}$                      | 0.2043075602291394                                   | exact corpus thermal weight |
| $H_0$                                        | $68.910590682276 \text{ km s}^{-1} \text{ Mpc}^{-1}$ | primary lock                |
| $a_0^{\text{BPB}}$                           | $1.367856592595 \times 10^{-10} \text{ m s}^{-2}$    | Unruh–dS balance            |

The key normalization point is the absence of a parasitic factor of  $y_t$  in the kinetic coefficient. In conformal time and in the notation of the HR-FLRW perturbation convention,

$$H_\omega = \frac{\xi'}{\xi} + H, \quad \delta H_\omega = \frac{\delta \xi'}{\xi} + H \frac{\delta \xi}{\xi}. \quad (166)$$

The Einstein–Hilbert kinetic contribution of the  $f$  metric contains

$$3M_{\text{Pl}}^2(\xi a)^2(\delta H_\omega)^2 = 3M_{\text{Pl}}^2(\xi a)^2 \frac{1}{\xi^2}(\delta \xi' + H \delta \xi)^2 = 3M_{\text{Pl}}^2 a^2(\delta \xi' + H \delta \xi)^2. \quad (167)$$

Thus the conformal-time kinetic normalization is

$$Z_y = 3M_{\text{Pl}}^2 a^2, \quad (168)$$

with no residual factor of  $y_t$ . In the corresponding cosmic-time action density this is the same normalization written with the usual extra factor of  $a$ .

The potential term gives

$$M_y^2 = \frac{m^2 V_g^{F''}(y_t)}{3M_{\text{Pl}}^2}, \quad H_0^2 = \frac{m^2 V_g^F(y_t)}{3M_{\text{Pl}}^2}, \quad (169)$$

and therefore Eq. (76). The branch algebra then yields  $\nu = 1/6$  and the Allen–Folacci factor  $f(\nu) = 27/2$  for the audited normalization. The closure-selected Fierz–Pauli scale is Eq. (78).

The rounded mode-energy fraction is defined by the virial ratio of the  $y$  oscillator in the BPB vacuum:

$$\sin^2 \theta_y^{\text{HR}} \equiv \frac{E_y}{E_{\text{branch}}} = \frac{m^2 V_g^{F''}(y_t) \langle \delta y^2 \rangle}{m^2 V_g^F(y_t)} = \frac{V_g^{F''}(y_t) \langle \delta y^2 \rangle}{V_g^F(y_t)}. \quad (170)$$

With  $\langle \delta y^2 \rangle = y_t^2$ , this gives

$$\sin^2 \theta_y^{\text{HR}} = \frac{V_g^{F''}(y_t) y_t^2}{V_g^F(y_t)} = \frac{\Delta \Omega}{V_g^F(y_t)} = 0.200. \quad (171)$$

The exact corpus thermal weight used for numerical application is instead

$$1 - s_{\text{surv},0} = 0.2043075602291394. \quad (172)$$

The difference is a recorded source-lock correction between the rounded HR branch algebra and the exact numerical corpus lock.

Finally, since the lower Bianchi point has  $H_f = 0$  in the reduced branch-flow interpretation,  $T_{\text{dS}}^{(f)} = 0$  and

$$T_{\text{eff}} = (1 - s_{\text{surv},0}) T_{\text{dS}}^{(g)}. \quad (173)$$

Equating  $T_{\text{eff}}$  to the local Unruh temperature gives Eq. (83). With the primary source locks this gives Eq. (85).

The status is therefore: D2 is closed as an internal, source-locked acceleration-scale bridge under the documented H1–H4 convention bundle. It should not be promoted, by itself, to a full nonrelativistic MOND action-level limit or to a full SPARC rotation-curve likelihood.

## References

- [1] Allen, B. and Folacci, A. (1985). Vacuum states in de Sitter space. *Physical Review D*, 32, 3136.
- [2] Comelli, D., Crisostomi, M., and Pilo, L. (2012). Perturbations in massive gravity cosmology. *Journal of High Energy Physics*, 2012(06), 085. arXiv:1202.1986.
- [3] Comelli, D., Crisostomi, M., and Pilo, L. (2014). FRW cosmological perturbations in massive bigravity. *Physical Review D*, 90, 084003. arXiv:1403.5679.
- [4] Boulware, D. G. and Deser, S. (1972). Can gravitation have a finite range? *Physical Review D*, 6, 3368–3382.
- [5] de Rham, C., Gabadadze, G., and Tolley, A. J. (2011). Resummation of massive gravity. *Physical Review Letters*, 106, 231101. arXiv:1011.1232.
- [6] Hassan, S. F. and Rosen, R. A. (2012a). Resolving the ghost problem in nonlinear massive gravity. *Physical Review Letters*, 108, 041101. arXiv:1106.3344.
- [7] Hassan, S. F. and Rosen, R. A. (2012b). Bimetric gravity from ghost-free massive gravity. *Journal of High Energy Physics*, 2012(2), 126. arXiv:1109.3515.
- [8] von Strauss, M., Schmidt-May, A., Enander, J., Mortsell, E., and Hassan, S. F. (2012). Cosmological solutions in bimetric gravity and their observational tests. *Journal of Cosmology and Astroparticle Physics*, 2012(03), 042. arXiv:1111.1655.